High-Dimensional Clustering Under Heterogeneous Structure

1 Problem Overview

Modern biological measurement technologies can record thousands of numerical measurements from individual cells or from small spatial regions of tissue. These measurements may represent gene-expression-like signals, protein-marker-like signals, morphology measurements, spatial coordinates, and local tissue-context summaries. A common scientific goal is to discover unknown biological structure from such data, such as cell types, activated cell states, rare subpopulations, developmental states, or spatial tissue niches.

This project studies unsupervised clustering under multiple high-dimensional data-generation settings that differ in structure and difficulty. Each row contains a feature vector, but no target label is provided to participants. The goal is to assign a cluster label

$$\hat{c}_i = f(x_i)$$

to each row so that the predicted clustering agrees with hidden ground-truth groups used only for evaluation.

Important note. No previous biological knowledge is required for this project. The biological description is only used to motivate the data-generating process. You should treat the task as a high-dimensional clustering problem involving noisy features, sparse informative signals, rare groups, hierarchical structure, and spatial context. Success depends on machine learning reasoning, not on domain-specific biology.

The benchmark is divided into four parts (A–D), each targeting a different clustering capability:

- Part A: sparse marker clustering in high-dimensional noisy data.
- Part B: rare cell-type discovery under severe cluster imbalance.
- Part C: hierarchical and continuous cell-state clustering.
- Part D: spatial tissue-niche clustering using both features and neighborhood context.

Participants may use either a single global clustering pipeline or separate part-specific pipelines selected using the `part` column. Both approaches are allowed.

2 Quick Start

This competition requires participants to assign a cluster label to every row in `test.csv`. The dataset is divided into four subproblems, identified by the column `part` $\in \{A, B, C, D\}$.

Depending on your modeling strategy, you may either cluster all rows using a single pipeline or use separate clustering methods for different parts. Since the parts have different objectives, part-specific pipelines are strongly encouraged.

For model development, use `train.csv` for exploratory analysis, preprocessing design, representation learning, and unsupervised model fitting. Participants then submit cluster labels for

`test.csv`. During the competition, the displayed leaderboard is computed on a public subset of the hidden test labels, while the final ranking is computed on the private remainder of the hidden test labels.
The main points to keep in mind are:

- This is an **unsupervised learning** task. No cluster labels are provided in the released files.
- The column `part` tells you which clustering subproblem a row belongs to.
- Rows from different parts should not be assumed to share the same cluster meanings.
- Cluster IDs are arbitrary. For example, your cluster ID `4` in Part A does not need to mean the same thing as cluster ID `4` in Part B.
- Parts A–D have different data geometry and may require different preprocessing or clustering methods.
- The number of clusters is not given exactly. You must estimate a reasonable clustering structure from the data.
- A simple method such as K-means on all raw features is not expected to perform well across all parts.

Performance is evaluated separately on Parts A–D using Adjusted Rand Index (ARI). The final competition score is computed as a weighted combination of the four part-wise ARI values, with higher values indicating better performance.
Your final submission should be a CSV file with exactly two columns:

```
row_id,cluster_id
```

The file must contain exactly one cluster assignment for each row in `test.csv`.

3 Files Provided

The release contains the following CSV files:

- `train.csv`: unlabeled training split for exploration and unsupervised fitting.
- `test.csv`: unlabeled test split used for leaderboard scoring and final evaluation.
- `sample_submission.csv`: required submission template.

Each of `train.csv` and `test.csv` is a single combined table containing rows from all four parts. The `part` column should be used to identify the subproblem associated with each row.

> Evaluation is reported separately for Parts A–D, but `test.csv` is a single combined table. Your submission must contain cluster assignments for *all* rows in `test.csv`.

> **Protocol and rules.**
>
> - **No released labels.** The released `train.csv` and `test.csv` files do not contain ground-truth cluster labels.
> - **Hidden test labels.** The hidden test labels are kept only by the evaluator and are

used only to compute leaderboard and final scores.

- **Preprocessing.** Fit scalers, imputers, feature-selection rules, dimensionality-reduction models, and other transformations using only allowed data.
- **Part-specific modeling.** You may build separate clustering pipelines for different values of the `part` column.
- **Cluster labels are arbitrary.** Evaluation is invariant to permutations of submitted cluster IDs.
- **Degenerate submissions.** Submissions assigning all rows to one cluster, or assigning nearly every row to its own cluster, are not meaningful and may receive very poor scores.

4 Dataset Schema

All four parts use a common tabular schema, but some columns are more meaningful for certain parts. In particular, the `part` column identifies the clustering subproblem, while the `mode` column indicates whether a row is a single-cell-style observation or a spatial tissue-spot observation:

- `part=A`: sparse marker clustering in high-dimensional noisy data.
- `part=B`: rare cell-type discovery under cluster imbalance.
- `part=C`: hierarchical and continuous cell-state clustering.
- `part=D`: spatial tissue-niche clustering using spatial context.
- `mode=single_cell`: an individual cell-like observation. These correspond to Parts A, B, and C.
- `mode=spatial_spot`: a small tissue-region-like observation with spatial context. These correspond to Part D.

For `single_cell` rows, spatial coordinates may be uninformative or set to placeholder values. For `spatial_spot` rows, `x_coord`, `y_coord`, and neighborhood-summary columns are meaningful and may help identify tissue niches.

5 Split Summary

The released files are combined tables containing rows from all four parts. The final test set contains rows with

$$\texttt{part} \in \{A, B, C, D\}.$$

The test set contains rows from all four subproblems:

$$\text{A: } N_A, \quad \text{B: } N_B, \quad \text{C: } N_C, \quad \text{D: } N_D.$$

The same test set can also be grouped by row type:

$$\texttt{single\_cell:} \quad N_{\text{single}}, \qquad \texttt{spatial\_spot:} \quad N_{\text{spatial}}.$$

The exact row counts are fixed in the released dataset. During the competition, leaderboard feedback is computed using a public subset of the hidden test labels, while the final ranking is based on the private remainder.

Column	Type	Description
`row_id`	int/string	Unique identifier used to align cluster assignments with rows.
`part`	str	Subproblem indicator: `A`, `B`, `C`, or `D`. Rows from different parts are generated from different clustering mechanisms and evaluated separately.
`mode`	str	Row type: `single_cell` or `spatial_spot`. Parts A–C use `single_cell`; Part D uses `spatial_spot`.
`g1..g500`	float	High-dimensional gene-expression-like features. These are noisy, sparse, and only partially informative.
`p1..p40`	float	Protein-marker-like features. These are lower-dimensional and may contain cluster-specific marker signals.
`morph1..morph30`	float	Morphology-like measurements representing shape, texture, size, or intensity summaries.
`qc_depth`	float	Quality-control feature representing measurement depth or total signal strength.
`qc_noise`	float	Quality-control feature representing measurement noise.
`dropout_rate`	float	Fraction-like feature indicating the amount of suppressed or missing signal.
`x_coord`	float	Spatial x-coordinate; primarily meaningful for `spatial_spot` rows in Part D. For Parts A–C this may be a placeholder or nuisance feature.
`y_coord`	float	Spatial y-coordinate; primarily meaningful for `spatial_spot` rows in Part D. For Parts A–C this may be a placeholder or nuisance feature.
`neighbor_density`	float	Local spatial-density summary; primarily meaningful in Part D.
`local_heterogeneity`	float	Neighborhood-diversity summary; primarily meaningful in Part D.
`batch_id`	category/int	Nuisance grouping variable that may affect measurements but is not itself the clustering target.
`cluster_id`	int	Hidden ground-truth cluster label used only by the evaluator. This column is **not** present in the released `train.csv` or `test.csv`.

Table 1: Dataset schema.

Single-cell rows (Parts A/B/C)

Parts A, B, and C contain high-dimensional single-cell-style observations. For these rows, `mode=single_cell`. The gene-like, protein-like, morphology, and quality-control features are meaningful. Spatial coordinates are either placeholders or weak nuisance variables and should not be assumed to define the hidden clusters.

Spatial tissue-spot rows (Part D)

Part D contains spatial tissue-spot observations. For these rows, `mode=spatial_spot`. In addition to molecular and morphology-like measurements, the coordinates `x_coord`, `y_coord`, and neighborhood-summary features may be important. The correct cluster may depend on both the measured feature values and the local spatial context.

- Part A: high-dimensional observations with sparse marker signals and many noisy dimensions.
- Part B: highly imbalanced observations containing both common and rare hidden groups.
- Part C: observations arranged around hierarchical and partially continuous state structure.

- Part D: spatial observations where tissue-niche identity depends on both features and neighborhood context.

Important clarification. The four parts are not different questions asked on the exact same rows. They are four different row subsets inside the same CSV files. Part A rows are evaluated only for the Part A clustering objective, Part B rows only for the Part B objective, and so on. Thus, Part A does not need to discover the rare clusters intentionally introduced in Part B.

6 Evaluation Parts (Modeling Requirements)

Part A: Sparse marker clustering

Goal: Identify hidden clusters that are defined by a small subset of informative marker features embedded in a much larger set of noisy features.

In this part, each row represents a single-cell-style observation. The hidden groups correspond to distinct cell-type-like populations. However, each population differs from the others only through a limited set of marker features among `g1..g500` and `p1..p40`. Many dimensions are weakly informative, redundant, or pure nuisance variation. As a result, distances computed directly in the full raw feature space may be unreliable.

A successful method should recognize that not all features are equally useful. Reasonable strategies may include normalization, variance filtering, feature selection, dimensionality reduction, or clustering in a learned low-dimensional representation. The main difficulty is to recover the meaningful marker-driven structure without being dominated by high-dimensional noise.

Main challenge: the true clusters are sparse in feature space; most dimensions are not directly useful.
Hint: try comparing clustering performance after feature filtering or dimensionality reduction rather than clustering all raw features directly.

Part B: Rare cell-type discovery

Goal: Discover small but meaningful hidden groups without confusing them with noise or merging them into nearby large clusters.

In this part, the data contain several common clusters and a few rare clusters. The rare clusters represent uncommon cell-type-like subpopulations or unusual biological states. These rare groups may contain only a small fraction of the observations, and they may lie close to larger populations in feature space. This makes them difficult to detect using methods that assume balanced, well-separated, or similarly sized clusters.

A simple centroid-based method may place cluster centers near the dominant groups and ignore small populations. On the other hand, overly aggressive outlier detection may incorrectly discard rare but valid clusters. The modeling challenge is to distinguish rare structure from random noise. Methods that preserve local neighborhoods or account for density variation may be useful.

Main challenge: rare clusters are small and close to larger groups, so they can be mistaken for outliers or absorbed into common clusters.
Hint: inspect local neighborhood structure and avoid relying only on global compactness or equal-size cluster assumptions.

Part C: Hierarchical and continuous cell states

Goal: Cluster observations that contain both discrete group structure and continuous transitions between related states.

In this part, the hidden groups correspond to cell-state-like populations. Some groups form broad families, while others represent finer sub-states within those families. In addition, some observations lie along gradual transition paths between states. Therefore, the structure is not simply a set of isolated spherical clusters. Nearby groups may share many features, and the boundary between two states may be gradual rather than sharp.

A method that uses too few clusters may merge meaningful sub-states, while a method that uses too many clusters may split continuous trajectories into artificial fragments. Graph-based clustering, spectral methods, manifold-aware embeddings, or stability-based model selection may be useful.

> **Main challenge:** the data contain hierarchical groups and continuous transitions, so the correct clustering is not obvious from global Euclidean distance alone.
> **Hint:** consider nearest-neighbor graphs, spectral embeddings, or other methods that can capture manifold-like structure before clustering.

Part D: Spatial tissue-niche clustering

Goal: Infer hidden tissue niches using both high-dimensional measurements and spatial neighborhood context.

In this part, each row represents a spatial tissue-spot-like observation. The hidden clusters correspond to tissue niches, which are regions defined not only by their own feature values but also by the surrounding spatial environment. Two spots may have similar expression-like features but belong to different niches because their neighboring spots are different. Conversely, two spots may have slightly different raw measurements but belong to the same niche if they occur in similar spatial contexts.

This part is the most context-dependent part of the benchmark. Treating the rows as independent tabular observations may miss important spatial organization. However, using spatial information too aggressively can also be harmful: excessive smoothing may erase rare niches or blur boundaries between neighboring regions. Participants should think carefully about how much spatial context to incorporate.

> **Main challenge:** cluster identity depends jointly on feature similarity and local spatial context.
> **Hint:** try constructing spatial-neighborhood summaries or nearest-neighbor graphs, but be careful not to over-smooth boundary or rare regions.

7 Cluster Definition and Evaluation Setup

Each row i belongs to exactly one part:

$$\texttt{part}_i \in \{A, B, C, D\}.$$

Within that part, the row has an unobserved ground-truth cluster label

$$c_i \in \{1, 2, \ldots, K_{\texttt{part}_i}\},$$

where $K_{\texttt{part}_i}$ is the hidden number of clusters for that part. Participants do not observe c_i for any row in the released files. Instead, participants submit a predicted cluster label $\hat{c}_i$.
For a row i in part p, the predicted cluster assignment may be written abstractly as

$$\hat{c}_i = f_p(x_i),$$

where x_i denotes the available feature vector and f_p is the clustering pipeline used for that part. A single global function f may also be used across all parts, but rows from different parts are still evaluated separately.

A clustering method may use the following information for row i:

$$\hat{c}_i = f_p\Big(g_i,\ p_i,\ \texttt{morph}_i,\ \texttt{qc\_depth}_i,\ \texttt{qc\_noise}_i,\ \texttt{dropout\_rate}_i,\\ \texttt{batch\_id}_i,\ \texttt{x\_coord}_i,\ \texttt{y\_coord}_i,\ \texttt{neighbor\_density}_i,\ \texttt{local\_heterogeneity}_i\Big). \tag{1}$$

For Part D, participants may additionally construct neighborhood-based features from the spatial coordinates and observed covariates.

Since ARI is computed separately within each part, the same submitted cluster ID may have different meanings in different parts. For example, cluster ID 2 in Part A and cluster ID 2 in Part B are not required to correspond to the same biological or statistical group.

8 Scoring

The evaluator computes Adjusted Rand Index (ARI) separately for each part and then forms a weighted aggregate. ARI compares two clusterings while accounting for agreement that may occur by chance. Since cluster names are arbitrary, ARI is suitable because it is invariant to permutations of cluster IDs.
Let T denote the set of all test rows, and let

$$T_p = \{i \in T : \texttt{part}_i = p\}$$

be the subset of test rows belonging to part p.

Per-part ARI:

$$\mathrm{ARI}_p = \mathrm{ARI}\left(\{c_i : i \in T_p\}, \{\hat{c}_i : i \in T_p\}\right), \qquad p \in \{A, B, C, D\}. \tag{2}$$

Final score (higher is better):

$$\mathrm{Score} = 0.20\,\mathrm{ARI}_A + 0.25\,\mathrm{ARI}_B + 0.30\,\mathrm{ARI}_C + 0.25\,\mathrm{ARI}_D. \tag{3}$$

A perfect clustering has score 1.0. A clustering that is no better than random assignment is expected to have ARI close to 0, although ARI can sometimes be negative for very poor clusterings.

Recommended cluster-count ranges

The exact number of hidden clusters is not provided. However, to prevent degenerate submissions, participants should keep the number of submitted clusters within broad reasonable ranges:

Part	Recommended submitted cluster range
A	5–15
B	6–20
C	8–25
D	8–30

These ranges are not intended to reveal the true number of clusters. They are provided only to discourage pathological solutions.

9 Submission Format

Submit a CSV with exactly two columns:

`row_id`, `cluster_id`.

Your submission must satisfy all of the following:

- every `row_id` from `test.csv` must appear exactly once;
- the predicted integer cluster label must be provided in the `cluster_id` column;
- the row order must remain unchanged from `test.csv`;
- cluster IDs must be non-negative integers;
- predictions must be provided for all rows from Parts A–D.

Example:

```
row_id,cluster_id
te_A_00001,4
te_B_00077,9
te_C_00152,2
te_D_00073,3
...
```

It is acceptable for the same integer cluster ID to appear in multiple parts. Since the evaluation is performed separately within each part, the label values are treated as arbitrary identifiers within that part.

10 Expected Content

A strong submission typically includes:

- a simple baseline clustering method;
- preprocessing choices for scaling, missing values, quality-control features, and nuisance variables;
- dimensionality-reduction or representation-learning choices;

- a strategy for estimating the number of clusters;
- a discussion of why different parts may require different clustering strategies;
- one or two lightweight ablations, such as:
  - raw features vs. dimensionality-reduced features;
  - with vs. without feature selection;
  - global clustering vs. part-specific clustering;
  - with vs. without spatial features for Part D;
  - different choices of the number of clusters.

11 Suggested Reading

The following resources may help you explore the broader context of clustering high-dimensional single-cell and spatial data. These readings are optional. The project can be completed using general machine learning ideas.

- Kiselev, V. Y., Andrews, T. S., and Hemberg, M. *Challenges in unsupervised clustering of single-cell RNA-seq data*. Nature Reviews Genetics, 2019.
  https://doi.org/10.1038/s41576-018-0088-9
- Traag, V. A., Waltman, L., and van Eck, N. J. *From Louvain to Leiden: guaranteeing well-connected communities*. Scientific Reports, 2019.
  https://doi.org/10.1038/s41598-019-41695-z
- McInnes, L., Healy, J., and Melville, J. *UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction*. 2018.
  https://arxiv.org/abs/1802.03426
- Ester, M., Kriegel, H.-P., Sander, J., and Xu, X. *A density-based algorithm for discovering clusters in large spatial databases with noise*. KDD, 1996.
  https://www.aaai.org/Papers/KDD/1996/KDD96-037.pdf
- scikit-learn documentation. *Clustering*.
  https://scikit-learn.org/stable/modules/clustering.html
- scikit-learn documentation. *Adjusted Rand Score*.
  https://scikit-learn.org/stable/modules/generated/sklearn.metrics.adjusted_rand_score.html